

Behavioral Customer Data Analysis & Customer Segmentation

Project Overview

This report analyzes more than 2,200 customers across store, web, and catalog channels. Using machine learning clustering techniques, customers were grouped into distinct and actionable buyer personas. The objective is to better understand customer behavior, improve marketing efficiency, and design targeted promotional strategies for each segment.

Customer Segments Analysis

Cluster 0: Mature Value-Seeking Families

Profile Summary

This segment consists of mature families with stable middle-to-high incomes. Although financially comfortable, they carefully manage their spending because of the high costs associated with raising older children. They are highly responsive to discounts and promotions.

Demographics and Financial Characteristics

- 100% parents.
- Family size typically ranges from 2 to 4 members.
- Households mainly include teenagers rather than young children.
- Customers in this segment have stable and relatively high annual incomes.

Shopping Behavior

- Balanced spending focused on essential family products, especially meat products.
- Lower spending on non-essential products such as sweets.
- Highest usage of deals.
- Preferred channels:
 1. Physical Stores
 2. Web Platforms
 3. Catalogs

Marketing Strategies

- Offer family-size bundles and bulk discount promotions.
- Send personalized digital coupons through SMS or mobile applications.
- Focus campaigns on weekly savings and essential household products.

Cluster 1: Young Budget Families

Profile Summary

This segment represents younger families in the early stages of family life. Due to limited income and childcare expenses, they are highly budget conscious but still spend emotionally on products related to their children.

Demographics and Financial Characteristics

- Younger parents compared to other family segments.
- Small households with approximately 2 to 3 members.
- Families include young children and rarely have teenagers.
- Lowest annual income and overall spending among all clusters.

Shopping Behavior

- Higher spending on sweets and fruits for children despite financial limitations.
- Highest number of website visits but relatively low online purchases.
- Websites are mainly used for price comparison and budgeting.
- Preferred channels:
 - Stores and websites
 - Catalogs are rarely used

Marketing Strategies

- Provide free shipping once purchases reach a minimum threshold.
- Create child-focused promotional bundles.
- Combine essential products with discounted children's items.

Cluster 2: Premium VIP Customers

Profile Summary

This segment represents the company's highest-value and most profitable customers. They prioritize quality, exclusivity, and premium experiences over dis-counts.

Demographics and Financial Characteristics

- 1% parents.
- Highest average annual income among all segments.
- Disposable income is directed toward luxury and personal lifestyle spending.

Shopping Behavior

- Highest spending on premium products such as wines and meats.
- Lowest website browsing activity but highest purchase values.
- Customers enter channels with strong purchasing intent.
- Preferred channels:
 1. Store
 2. Catalog

Marketing Strategies

- Develop exclusive VIP membership programs.
- Focus on premium experiences instead of discount campaigns.
- Offer early access to luxury products and personalized services.
- Avoid mass-market coupon strategies.

Cluster 3: Financially Pressured Large Families

Profile Summary

This segment includes large families experiencing strong financial pressure. Although their income is slightly higher than Cluster 1, their spending remains limited because household expenses are distributed across many family members.

Demographics and Financial Characteristics

- 100% parents.
- Largest household sizes among all clusters, usually 4 to 5 members.
- Families contain both young children and teenagers.
- Mainly middle-aged parents aged 40 and above.

Shopping Behavior

- Clear gap between income and spending due to family responsibilities.
- Highly price-sensitive and focused on budgeting.
- Frequently visit the website but make relatively few purchases.
- Prefer physical store shopping.

Marketing Strategies

- Introduce bulk-buy promotions and family-size packaging.
- Offer discounts on essential products and frozen foods.
- Design mixed-age family bundles for both teenagers and young children.